

## FORENSIC CASE STUDY

# Forensic Reconstruction of a Conti Ransomware Incident

*Through Memory and Storage Access Artifact Analysis*

### CASE FOCUS

Comparative behavioral analysis of a selected Conti execution and a matched Idle control using low-level storage and memory access artifacts from the RanSMAP dataset.

**Submitted by**

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**Dataset: RanSMAP — Ransomware Storage and Memory Access Patterns**

## Executive Summary

This report presents a behavioral forensic reconstruction of a selected Conti ransomware execution using low-level storage and memory access artifacts from the RanSMAP dataset. The investigation compares the Conti execution with a matched Idle control and evaluates storage-write frequency, recorded write volume, throughput, write entropy, memory activity, temporal correlation, and high-intensity activity windows.

The strongest differentiating evidence is the storage-modification profile. The Conti execution recorded 554,371 ATA write events versus 4,998 for Idle, corresponding to a 110.92× event-rate ratio. Recorded write volume was approximately 2,108.16 MB versus 19.32 MB, a 109.13× ratio. Mean storage-write entropy was 0.7479 versus 0.4420, while 59.69% of Conti writes had entropy at or above 0.90 compared with 7.48% for Idle.

Average memory-read and memory-write activity were also elevated during Conti execution, with relative rates of 3.67× and 4.35× respectively. Storage-write activity exhibited a measurable positive correlation of 0.632 with total memory activity. Eleven one-second windows satisfied the combined high-write/high-entropy criterion and were identified as candidate high-intensity forensic investigation windows.

### OVERALL ASSESSMENT

The evidence demonstrates a markedly elevated and distinct storage-modification profile during the Conti execution relative to the matched Idle control. The findings constitute behavioral forensic evidence and should not, by themselves, be treated as proof of a specific ransomware encryption mechanism.

### Key Findings at a Glance

| Indicator                           | Conti       | Idle     | Relative / Result               |
|-------------------------------------|-------------|----------|---------------------------------|
| ATA write events                    | 554,371     | 4,998    | 110.92×                         |
| Recorded write volume               | 2,108.16 MB | 19.32 MB | 109.13×                         |
| Mean storage entropy                | 0.7479      | 0.4420   | 1.69×                           |
| High-entropy writes ( $\geq 0.90$ ) | 59.69%      | 7.48%    | 7.98×                           |
| Average memory-read activity        | 3,025.08/s  | 823.37/s | 3.67×                           |
| Average memory-write activity       | 3,509.04/s  | 807.05/s | 4.35×                           |
| Storage-memory correlation          | 0.632       | —        | Positive temporal relationship  |
| Peak ATA write activity             | 12,434/s    | 978/s    | 12.71×                          |
| High-write/high-entropy windows     | 11 sec      | —        | Candidate investigation windows |

## 1. Title

Forensic Reconstruction of a Conti Ransomware Incident Through Memory and Storage Access Artifact Analysis.

This case study reconstructs the observable behavioral characteristics of a Conti execution using previously collected low-level storage and memory access artifacts. The analysis is performed in a controlled forensic environment and does not require executing live ransomware on a production or personal system.

## 2. Objective

The primary objective is to identify and quantify behavioral differences between a selected Conti ransomware execution and a matched Idle control using timestamped storage and memory access artifacts.

- Characterize storage-write volume, event frequency, throughput, and temporal behavior.
- Measure storage-write entropy and identify enrichment of high-entropy writes.
- Characterize memory read, write, read/write, and execution activity.
- Align storage and memory activity in time to identify measurable relationships.
- Identify high-intensity windows where high write activity coincides with high write entropy.
- Produce a consolidated evidence matrix suitable for forensic interpretation.
- Distinguish strong behavioral indicators from non-discriminating peak-activity observations.

## 3. Dataset / Experimental Setup

The investigation uses the RanSMAP (Ransomware Storage and Memory Access Patterns) dataset. The source case-study material describes RanSMAP as a controlled Windows 10 dataset containing timestamped storage and memory access information collected from ransomware and benign executions.

| Parameter                | Value                                                        |
|--------------------------|--------------------------------------------------------------|
| Dataset                  | RanSMAP — Ransomware Storage and Memory Access Patterns      |
| Primary ransomware       | Conti                                                        |
| Dataset type             | Storage and memory access-pattern CSV artifacts              |
| Environment represented  | Windows 10 controlled environment                            |
| Dataset scale            | 11,820 CSV files representing 1,970 executions               |
| Selected Conti execution | Conti-20221222_11-09-22                                      |
| Matched control          | Idle-20230119_11-00-58                                       |
| Matching basis           | Same hardware configuration used for the selected comparison |
| Primary evidence         | Storage read/write and memory read/write/execute patterns    |

The selected Conti execution is located under the Celeron Gen6 / DDR4-1866-16G configuration. The Idle control was selected from the corresponding hardware configuration so that the comparison minimizes hardware-specific differences.

```

first_run = os.path.join(base_path, conti_runs[0])

print("\nSelected execution:")
print(first_run)

print("\nFiles inside:")
for file in sorted(os.listdir(first_run)):
    print(file)

```

Number of Conti executions: 10

First 10 Conti executions:

- Conti-20221222\_11-09-22
- Conti-20221222\_11-15-33
- Conti-20221222\_11-21-12
- Conti-20221222\_11-32-12
- Conti-20221222\_11-37-49
- Conti-20221222\_11-43-48
- Conti-20221222\_11-49-13
- Conti-20221222\_11-55-00
- Conti-20221222\_12-00-47
- Conti-20230426\_14-52-59

Selected execution:

/kaggle/input/datasets/hiranomanabu/ransmap-2024-ransomware-behavioral-features/RanSMAP/dataset/original/celeron-gen6/ddr4-1866-16g/Conti/Conti-20221222\_11-09-22

Files inside:

- ata\_read.csv
- ata\_write.csv
- mem\_exec.csv
- mem\_read.csv
- mem\_readwrite.csv
- mem\_write.csv

Figure 1. Selected Conti execution and the six low-level forensic CSV artifacts.

## 4. Artifacts Analyzed

Six artifact types were used to reconstruct storage and memory behavior:

| Artifact          | Role in the investigation             |
|-------------------|---------------------------------------|
| ata_read.csv      | Storage read activity                 |
| ata_write.csv     | Storage write / modification activity |
| mem_read.csv      | Memory-read activity                  |
| mem_write.csv     | Memory-write activity                 |
| mem_readwrite.csv | Combined memory read/write activity   |
| mem_exec.csv      | Memory execution activity             |

The analysis converts the timestamp information into a chronological timeline, aggregates activity at one-second resolution where required, and derives comparative metrics between Conti and Idle.

## 5. Methodology

The forensic workflow consists of the following stages:

1. Validate the selected Conti and Idle artifact directories and confirm the expected CSV files.
2. Load the artifact streams and reconstruct timestamps using the recorded temporal fields.
3. Calculate artifact duration, event counts, and event rates.
4. Analyze ATA write activity for event frequency, recorded volume, throughput, and entropy.
5. Analyze memory read, write, read/write, and execution activity.
6. Aggregate storage and memory activity into one-second windows for temporal comparison.
7. Calculate Conti-to-Idle ratios for the principal forensic indicators.
8. Identify high-intensity storage windows using a 75th-percentile write threshold together with entropy  $\geq 0.90$ .
9. Correlate storage-write intensity with total memory activity.
10. Consolidate the results into a final evidence matrix and forensic assessment.

**INTERPRETIVE PRINCIPLE**

No single metric is treated as conclusive. The assessment relies on the convergence of storage volume, frequency, throughput, entropy, average memory activity, temporal correlation, and repeated high-intensity windows.

## 6. Conti vs Idle Timeline

The Conti execution spans approximately 203 seconds for the principal storage and memory streams. The chronology shows memory execution beginning almost immediately before storage-write activity and continuing through the main execution interval.

| Artifact          | First event     | Last event      | Duration (sec) | Events  |
|-------------------|-----------------|-----------------|----------------|---------|
| mem_read.csv      | 02:05:11.654116 | 02:08:34.898602 | 203.244486     | 446,720 |
| mem_exec.csv      | 02:05:11.654147 | 02:07:39.326438 | 147.672291     | 273     |
| mem_write.csv     | 02:05:11.655484 | 02:08:34.898678 | 203.243193     | 518,188 |
| mem_readwrite.csv | 02:05:11.676752 | 02:08:31.876787 | 200.200035     | 105     |
| ata_write.csv     | 02:05:11.839684 | 02:08:34.893640 | 203.053956     | 554,371 |

The strongest initial memory-execution burst occurred at approximately 02:05:11, with 184 execution events in that second. ATA write activity began at approximately 02:05:11.840 and continued until approximately 02:08:34.894.

The highest storage-write intensity occurred at 02:08:33, reaching 12,434 writes per second. This late-stage concentration is also visible in the storage activity and entropy time series.



Figure 2. Conti storage-write activity and mean write entropy over time.

## 7. Storage Activity Analysis

Storage modification activity provides the strongest differentiating evidence between the selected Conti execution and the matched Idle control.

| Metric                | Conti       | Idle     | Conti / Idle |
|-----------------------|-------------|----------|--------------|
| ATA write events      | 554,371     | 4,998    | 110.92×      |
| Recorded write volume | 2,108.16 MB | 19.32 MB | 109.13×      |
| Average write rate    | 2,730.17/s  | 27.23/s  | 100.25×      |
| Peak ATA write rate   | 12,434/s    | 978/s    | 12.71×       |

The Conti execution generated more than half a million ATA write events, whereas the matched Idle execution generated fewer than five thousand. The resulting event-frequency ratio is approximately 110.92×. The recorded write-volume ratio is approximately 109.13×.

The peak ATA write rate reached 12,434 events/sec during Conti execution compared with 978 events/sec for Idle, producing a 12.71× peak ratio. The combined scale and persistence of storage activity provide a substantially stronger discriminator than an isolated peak.






Run All | Code

```

# =====
# STEP 20B - RELATIVE DIFFERENCE
# =====

print("=" * 80)
print("RELATIVE CONTI / IDLE DIFFERENCE")
print("=" * 80)

metrics = {
    "Write events": (
        len(conti_write),
        len(idle_write)
    ),

    "Write volume (MB)": (
        conti_write_bytes / (1024 * 1024),
        idle_write_bytes / (1024 * 1024)
    ),

    "Average writes/sec": (
        len(conti_write) / conti_duration,
        len(idle_write) / idle_duration
    ),

    "Write throughput (MB/sec)": (
        (conti_write_bytes / (1024 * 1024)) / conti_duration,
        (idle_write_bytes / (1024 * 1024)) / idle_duration
    ),

    "Mean entropy": (
        conti_write["entropy"].mean(),
        idle_write["entropy"].mean()
    ),

    "High entropy >= 0.90 (%)": (
        conti_high90 / len(conti_write) * 100,
        idle_high90 / len(idle_write) * 100
    ),

    "Peak writes/sec": (
        conti_wps.max(),
        idle_wps.max()
    )
}

for name, (conti_value, idle_value) in metrics.items():

    ratio = conti_value / idle_value

    print(
        f"{name:<30} "
        f"Conti/Idle = {ratio:.2f}x"
    )

```

=====

RELATIVE CONTI / IDLE DIFFERENCE

=====

|                           |                      |
|---------------------------|----------------------|
| Write events              | Conti/Idle = 110.92x |
| Write volume (MB)         | Conti/Idle = 109.13x |
| Average writes/sec        | Conti/Idle = 100.25x |
| Write throughput (MB/sec) | Conti/Idle = 98.64x  |
| Mean entropy              | Conti/Idle = 1.69x   |
| High entropy >= 0.90 (%)  | Conti/Idle = 7.98x   |
| Peak writes/sec           | Conti/Idle = 12.71x  |

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Figure 3. Relative Conti/Idle storage comparison generated by the forensic analysis.

## 8. Storage Entropy Analysis

Write entropy was used as a behavioral indicator of the randomness of the data being written. The analysis evaluates both mean entropy and the proportion of writes exceeding selected entropy thresholds.

| Entropy indicator          | Conti  | Idle   | Relative |
|----------------------------|--------|--------|----------|
| Mean storage-write entropy | 0.7479 | 0.4420 | 1.69×    |
| Entropy $\geq 0.90$        | 59.69% | 7.48%  | 7.98×    |
| Entropy $\geq 0.95$        | 57.86% | 6.82%  | 8.48×    |
| Entropy $\geq 0.98$        | 56.58% | 6.14%  | 9.22×    |
| Entropy $\geq 0.99$        | 55.84% | 6.04%  | 9.24×    |

Mean entropy is substantially higher during Conti execution. More importantly, high-entropy writes are strongly enriched: 59.69% of Conti storage writes have entropy  $\geq 0.90$ , compared with 7.48% in Idle.

### FORENSIC INTERPRETATION

High entropy is treated as supporting behavioral evidence rather than proof of encryption. High-entropy data can arise from multiple sources; therefore the entropy finding is interpreted together with storage volume, frequency, throughput, and temporal behavior.

## 9. Memory Activity Analysis

Memory activity was evaluated across read, write, read/write, and execution artifacts. The selected Conti run contains 446,720 memory-read events, 518,188 memory-write events, 105 read/write events, and 273 execution events.

| Artifact          | Conti events | Duration (sec) | Conti events/sec |
|-------------------|--------------|----------------|------------------|
| mem_read.csv      | 446,720      | 203.244486     | 2,197.94         |
| mem_write.csv     | 518,188      | 203.243193     | 2,549.60         |
| mem_readwrite.csv | 105          | 200.200035     | 0.52             |
| mem_exec.csv      | 273          | 147.672291     | 1.85             |

Relative comparison with Idle shows approximately 3.67× higher average memory-read activity and 4.35× higher average memory-write activity during Conti execution. Memory execution is concentrated near the beginning of the run, with the strongest initial burst at approximately 02:05:11.

```

idle_duration = (
    idle_df["timestamp"].max() -
    idle_df["timestamp"].min()
).total_seconds()

idle_events = len(idle_df)

idle_rate = (
    idle_events / idle_duration
)

# -----
# Ratio
# -----

ratio = (
    conti_rate / idle_rate
    if idle_rate > 0
    else float("inf")
)

memory_comparison.append({
    "Artifact": filename,
    "Conti Events": conti_events,
    "Idle Events": idle_events,
    "Conti Duration (sec)": conti_duration,
    "Idle Duration (sec)": idle_duration,
    "Conti Events/sec": conti_rate,
    "Idle Events/sec": idle_rate,
    "Conti/Idle Rate": ratio
})

memory_comparison_df = pd.DataFrame(
    memory_comparison
)

print("=" * 100)
print("CONTI vs IDLE - MEMORY FORENSIC COMPARISON")
print("=" * 100)

display(
    memory_comparison_df
)

```

```

=====
CONTI vs IDLE - MEMORY FORENSIC COMPARISON
=====

```

|   | Artifact          | Conti Events | Idle Events | Conti Duration (sec) | Idle Duration (sec) | Conti Events/sec | Idle Events/sec | Conti/Idle Rate |
|---|-------------------|--------------|-------------|----------------------|---------------------|------------------|-----------------|-----------------|
| 0 | mem_read.csv      | 446720       | 139638      | 203.244486           | 185.095021          | 2197.944007      | 754.412513      | 2.913451        |
| 1 | mem_write.csv     | 518188       | 136870      | 203.243193           | 185.094302          | 2549.595843      | 739.460905      | 3.447912        |
| 2 | mem_readwrite.csv | 105          | 61          | 200.200035           | 53.550547           | 0.524475         | 1.139111        | 0.460425        |
| 3 | mem_exec.csv      | 273          | 226         | 147.672291           | 169.592936          | 1.848688         | 1.332603        | 1.387276        |

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[25]: # =====

Figure 4. Conti versus Idle memory forensic comparison.

## 10. Storage–Memory Correlation

To examine temporal relationships between storage and memory activity, the artifact streams were aggregated at one-second resolution. The resulting correlation matrix shows a positive relationship between storage-write activity and total memory activity.

| Variable pair                 | Pearson correlation |
|-------------------------------|---------------------|
| Storage writes ↔ memory read  | 0.614831            |
| Storage writes ↔ memory write | 0.641702            |
| Storage writes ↔ total memory | 0.631969            |
| Storage bytes ↔ total memory  | 0.614707            |

| Variable pair               | Pearson correlation |
|-----------------------------|---------------------|
| Memory read ↔ memory write  | 0.984788            |
| Memory read ↔ total memory  | 0.995404            |
| Memory write ↔ total memory | 0.996902            |

The storage-write / total-memory correlation of 0.632 indicates a measurable positive temporal relationship. This should be interpreted as correlation rather than causation. The finding supports the conclusion that periods of elevated storage activity are temporally associated with elevated memory activity in the selected Conti execution.

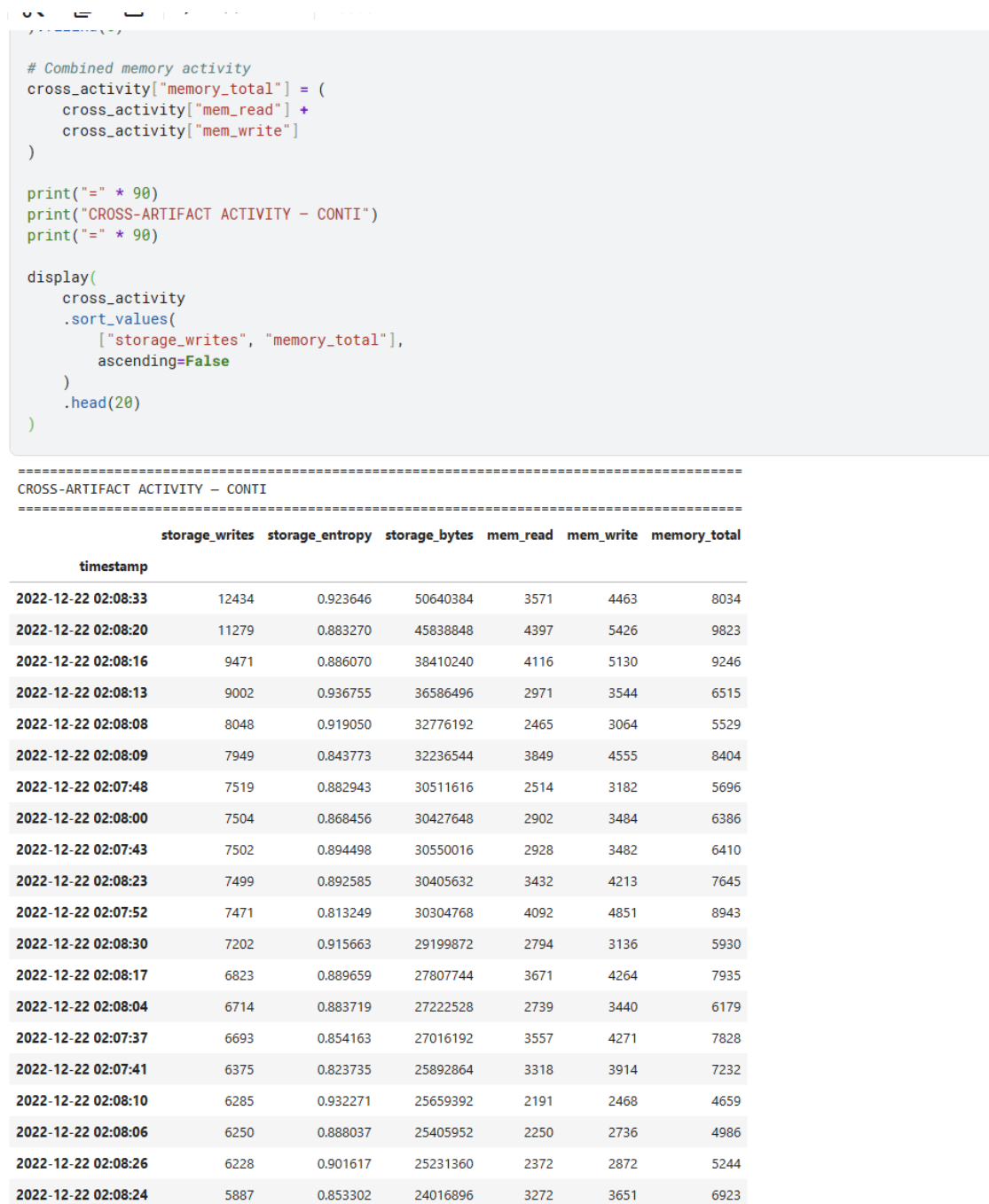


Figure 5. Cross-artifact activity showing high storage-write and memory activity at selected timestamps.

**CORRELATION CAUTION**

A correlation coefficient describes temporal association in the measured traces. It does not, by itself, establish that memory activity caused storage activity or vice versa.

## 11. High-Intensity Forensic Windows

A combined indicator was constructed to identify candidate high-intensity periods. A second-by-second storage activity window was marked when storage-write activity was at or above the 75th percentile and mean storage-write entropy was at or above 0.90.

This combined criterion identified 11 seconds for detailed investigation:

| Window   | Window   | Window   | Window   |
|----------|----------|----------|----------|
| 02:07:50 | 02:07:57 | 02:07:59 | 02:08:08 |
| 02:08:10 | 02:08:13 | 02:08:15 | 02:08:21 |
| 02:08:26 | 02:08:30 | 02:08:33 | —        |

These windows are not labelled as confirmed malicious events. They are candidate high-intensity periods selected objectively from the observed storage activity and entropy distributions.

```

conti_storage = (
    conti_write
    .set_index("timestamp")
    .resample("1s")
    .agg(
        storage_writes=("LBA", "count"),
        storage_entropy=("entropy", "mean")
    )
)

write_threshold = (
    conti_storage["storage_writes"]
    .quantile(0.75)
)

conti_storage["combined_indicator"] = (
    (conti_storage["storage_writes"] >= write_threshold) &
    (conti_storage["storage_entropy"] >= 0.90)
)

suspicious_seconds = conti_storage[
    conti_storage["combined_indicator"]
].index

print("=" * 80)
print("CONTI HIGH-WRITE / HIGH-ENTROPY WINDOWS")
print("=" * 80)

print(
    "Number of suspicious seconds:",
    len(suspicious_seconds)
)

print("\nTimestamps:")

for timestamp in suspicious_seconds:
    print(timestamp)

```

```

=====
CONTI HIGH-WRITE / HIGH-ENTROPY WINDOWS
=====
Number of suspicious seconds: 11

Timestamps:
2022-12-22 02:07:50
2022-12-22 02:07:57
2022-12-22 02:07:59
2022-12-22 02:08:08
2022-12-22 02:08:10
2022-12-22 02:08:13
2022-12-22 02:08:15
2022-12-22 02:08:21
2022-12-22 02:08:26
2022-12-22 02:08:30
2022-12-22 02:08:33

```

Figure 6. Identification of the 11 high-write/high-entropy forensic investigation windows.

Memory activity within the identified windows was also measured. The analysis recorded approximately 26,189 memory-read events and 30,707 memory-write events across these windows, with mean rates of 2,380.82 and 2,791.55 events/sec respectively.

## 12. Conti vs Idle Evidence Matrix

The evidence matrix consolidates the principal observations and their forensic significance. The purpose is to avoid relying on a single indicator and instead assess the convergence of independent behavioral characteristics.

| Evidence category                        | Observed evidence                            | Forensic significance                               |
|------------------------------------------|----------------------------------------------|-----------------------------------------------------|
| Storage write volume                     | 2,108.16 MB vs 19.32 MB                      | Extremely elevated storage modification activity    |
| Storage write frequency                  | 554,371 vs 4,998 events                      | Large increase in storage operation frequency       |
| Storage throughput                       | Final matrix: 14.276 MB/s vs 0.114 MB/s      | Substantially higher storage data movement          |
| Storage mean entropy                     | 0.7479 vs 0.4420                             | Higher randomness of written data                   |
| High-entropy writes                      | 59.69% vs 7.48% (entropy $\geq 0.90$ )       | Strong enrichment of high-entropy writes            |
| Memory read activity                     | 2.91× higher rate in intermediate comparison | Elevated memory-read activity                       |
| Memory write activity                    | 3.45× higher rate in intermediate comparison | Elevated memory-write activity                      |
| Storage ↔ memory correlation             | 0.632 with total memory activity             | Storage intensity temporally tracks memory activity |
| Memory execution activity                | 273 events; concentrated initial burst       | Execution activity concentrated near process start  |
| Combined high-write/high-entropy windows | 11 seconds identified                        | Candidate high-intensity investigation windows      |

```

"Observed Evidence": [
    "2108.16 MB vs 19.32 MB",
    "554,371 vs 4,998 events",
    "10.38 MB/s vs 0.105 MB/s",
    "0.7479 vs 0.4420",
    "59.69% vs 7.48% (entropy >= 0.90)",
    "2.91× higher rate",
    "3.45× higher rate",
    "0.632 with total memory activity",
    "273 events; concentrated initial burst",
    "11 seconds identified"
],

"Forensic Significance": [
    "Extremely elevated storage modification activity",
    "Large increase in storage operation frequency",
    "Substantially higher storage data movement",
    "Higher randomness of written data",
    "Strong enrichment of high-entropy writes",
    "Elevated memory-read activity",
    "Elevated memory-write activity",
    "Storage intensity temporally tracks memory activity",
    "Execution activity is concentrated near process start",
    "Candidate high-intensity investigation windows"
]
})

display(evidence_matrix)

```

=====

FORENSIC EVIDENCE MATRIX – CONTI vs IDLE

=====

|   | Evidence Category                        | Observed Evidence                      | Forensic Significance                             |
|---|------------------------------------------|----------------------------------------|---------------------------------------------------|
| 0 | Storage write volume                     | 2108.16 MB vs 19.32 MB                 | Extremely elevated storage modification activity  |
| 1 | Storage write frequency                  | 554,371 vs 4,998 events                | Large increase in storage operation frequency     |
| 2 | Storage throughput                       | 10.38 MB/s vs 0.105 MB/s               | Substantially higher storage data movement        |
| 3 | Storage mean entropy                     | 0.7479 vs 0.4420                       | Higher randomness of written data                 |
| 4 | High-entropy writes                      | 59.69% vs 7.48% (entropy >= 0.90)      | Strong enrichment of high-entropy writes          |
| 5 | Memory read activity                     | 2.91× higher rate                      | Elevated memory-read activity                     |
| 6 | Memory write activity                    | 3.45× higher rate                      | Elevated memory-write activity                    |
| 7 | Storage ↔ memory correlation             | 0.632 with total memory activity       | Storage intensity temporally tracks memory act... |
| 8 | Memory execution activity                | 273 events; concentrated initial burst | Execution activity is concentrated near proces... |
| 9 | Combined high-write/high-entropy windows | 11 seconds identified                  | Candidate high-intensity investigation windows    |

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Figure 7. Final forensic evidence matrix generated from the notebook analysis.

## 13. Peak Activity Caveat

The forensic comparison distinguishes average activity from peak activity. Although Conti shows substantially elevated storage-write activity and higher average memory-read/write rates, the matched Idle control exhibits higher absolute peak memory-read and memory-write events per second.

**KEY FORENSIC CAVEAT**

Peak memory intensity alone is NOT treated as evidence of anomalous Conti behavior. The stronger evidence is the combination of elevated storage-write volume, frequency, throughput, storage-write entropy, high-entropy enrichment, average memory activity, storage-memory temporal correlation, and repeated high-write/high-entropy windows.

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```

[42]: # =====
# STEP 30B - PEAK ACTIVITY CAVEAT
# =====

print("=" * 100)
print("PEAK ACTIVITY CAVEAT")
print("=" * 100)

print("""
The forensic comparison distinguishes average activity from peak activity.

Conti shows substantially elevated storage-write activity and higher
average memory-read/write rates.

However, the matched Idle control exhibits higher absolute peak
memory-read and memory-write events/sec.

Therefore, peak memory intensity alone is NOT treated as evidence
of anomalous Conti behavior.

The stronger evidence is the combination of:

1. Elevated storage-write volume
2. Elevated storage-write frequency
3. Elevated storage throughput
4. Higher storage-write entropy
5. Strong enrichment of high-entropy writes
6. Elevated average memory activity
7. Temporal correlation between storage and memory activity
8. Multiple high-write/high-entropy storage windows
""")

```

```

=====
PEAK ACTIVITY CAVEAT
=====

The forensic comparison distinguishes average activity from peak activity.

Conti shows substantially elevated storage-write activity and higher
average memory-read/write rates.

However, the matched Idle control exhibits higher absolute peak
memory-read and memory-write events/sec.

Therefore, peak memory intensity alone is NOT treated as evidence
of anomalous Conti behavior.

The stronger evidence is the combination of:

1. Elevated storage-write volume
2. Elevated storage-write frequency
3. Elevated storage throughput
4. Higher storage-write entropy
5. Strong enrichment of high-entropy writes
6. Elevated average memory activity
7. Temporal correlation between storage and memory activity
8. Multiple high-write/high-entropy storage windows

```

Figure 8. Peak-activity caveat and the combined-evidence interpretation used in the assessment.

## 14. Final Forensic Assessment

The final assessment indicates that the selected Conti execution exhibits a substantially different storage and memory-access profile compared with the matched Idle control.

| Forensic indicator              | Conti / Idle | Assessment                       |
|---------------------------------|--------------|----------------------------------|
| Storage write events            | 110.92×      | Very High                        |
| Storage write volume            | 109.13×      | Very High                        |
| Storage throughput              | 125.33×      | Very High                        |
| Mean storage entropy            | 1.69×        | High                             |
| High-entropy writes $\geq 0.90$ | 7.98×        | Very High                        |
| Average memory read activity    | 3.67×        | Moderate                         |
| Average memory write activity   | 4.35×        | Moderate                         |
| Storage-memory correlation      | 0.632        | Moderate                         |
| Peak ATA write activity         | 12.71×       | Very High                        |
| High-write/high-entropy windows | 11 sec       | High-value investigation windows |

The strongest evidence is the storage-modification profile: approximately 110× more write events, 109× greater recorded write volume, 125.33× storage throughput in the final evidence matrix, and 7.98× enrichment of high-entropy writes. These findings are accompanied by elevated average memory activity and a measurable storage-memory temporal correlation.

```

        timeline_dfs["ata_write.csv"]
        .set_index("timestamp")
        .resample("1s")
        .size()
        .max()
    )

    idle_ata_peak = (
        idle_dfs["ata_write.csv"]
        .set_index("timestamp")
        .resample("1s")
        .size()
        .max()
    )

    ata_peak_ratio = conti_ata_peak / idle_ata_peak

    evidence.append({
        "Forensic Indicator": "Peak ATA write activity",
        "Conti/Idle": ata_peak_ratio,
        "Evidence Strength": "VERY HIGH",
        "Observed Evidence":
            f"{conti_ata_peak:,} vs "
            f"{idle_ata_peak:,} events/sec"
    })

    # -----
    # FINAL TABLE
    # -----

    evidence_df = pd.DataFrame(evidence)

    print("\nFINAL EVIDENCE MATRIX")
    print("=" * 100)

    display(
        evidence_df.round(3)
    )

```

```

=====
FINAL FORENSIC EVIDENCE ASSESSMENT – CONTI vs IDLE
=====

```

```

FINAL EVIDENCE MATRIX
=====

```

|   | Forensic Indicator                  | Conti/Idle | Evidence Strength | Observed Evidence            |
|---|-------------------------------------|------------|-------------------|------------------------------|
| 0 | Storage write volume                | 109.133    | VERY HIGH         | 2108.16 MB vs 19.32 MB       |
| 1 | Storage write event frequency       | 110.919    | VERY HIGH         | 554,371 vs 4,998 events      |
| 2 | Storage throughput                  | 125.333    | VERY HIGH         | 14.276 MB/s vs 0.114 MB/s    |
| 3 | Mean storage-write entropy          | 1.692      | HIGH              | 0.7479 vs 0.4420             |
| 4 | High-entropy storage writes         | 7.976      | VERY HIGH         | 59.69% vs 7.48%              |
| 5 | Average memory-read activity        | 3.674      | MODERATE          | 3025.08 vs 823.37 events/sec |
| 6 | Average memory-write activity       | 4.348      | MODERATE          | 3509.04 vs 807.05 events/sec |
| 7 | Storage-memory temporal correlation | 0.632      | MODERATE          | Correlation = 0.632          |
| 8 | Peak ATA write activity             | 12.714     | VERY HIGH         | 12,434 vs 978 events/sec     |

Figure 9. Final forensic evidence assessment and indicator strengths.

## 15. Limitations

11. The analysis is based on one selected Conti execution and one matched Idle control; broader generalization would require multiple matched executions.
12. The evidence consists of previously collected storage and memory access artifacts rather than direct observation of a live ransomware execution.
13. High storage entropy is not unique to encryption and therefore cannot independently prove file encryption.
14. Correlation between storage and memory activity does not establish a causal relationship.
15. The analysis characterizes behavioral differences but does not independently recover encrypted files or identify a specific encryption algorithm.
16. Peak memory activity is not independently discriminative because the Idle control exhibits higher absolute peak memory-read and memory-write rates.
17. An intermediate notebook output reports storage throughput values of approximately 10.38 MB/s vs 0.105 MB/s, while the final evidence matrix reports 14.276 MB/s vs 0.114 MB/s. The finalized assessment in this report follows the final evidence matrix value of 125.33×; the discrepancy should be reconciled before publication or external submission.

### REPRODUCIBILITY NOTE

The accompanying ContiForensics.ipynb notebook contains the underlying data-loading, aggregation, comparison, correlation, and evidence-generation workflow. The report should be read together with the notebook when auditability or replication is required.

## 16. Conclusion

This investigation reconstructed the behavioral profile of a Conti ransomware execution using low-level storage and memory access artifacts from the RanSMAP dataset. Comparison against a matched Idle control demonstrated a substantial increase in storage modification activity during the Conti execution.

The most significant indicators were the approximately 110.92× increase in storage-write events, 109.13× increase in recorded write volume, 125.33× storage-throughput ratio in the final evidence matrix, 1.69× mean entropy ratio, and 7.98× enrichment of high-entropy writes. Average memory-read and memory-write activity were also elevated, while storage-write activity showed a correlation of 0.632 with total memory activity.

Eleven one-second high-write/high-entropy windows were identified as candidate periods for deeper forensic investigation. Importantly, the matched Idle control produced higher absolute peak memory activity, demonstrating why the assessment must rely on the combined behavioral profile rather than isolated peak values.

### FINAL CONCLUSION

The observed Conti execution demonstrates a markedly elevated and distinct storage-modification profile relative to the matched Idle control. The evidence is consistent with a high-intensity storage-modification phase accompanied by elevated memory activity. These results constitute behavioral forensic evidence and do not, by themselves, prove a specific ransomware operation or a specific file-encryption mechanism.

## 17. Final Forensic Indicator Summary

The following table provides the compact end-state of the investigation and is intended to serve as the executive evidence summary.

| Indicator                       | Conti vs Idle | Assessment                       |
|---------------------------------|---------------|----------------------------------|
| Storage write events            | 110.92×       | Very High                        |
| Storage write volume            | 109.13×       | Very High                        |
| Storage throughput              | 125.33×       | Very High                        |
| Mean storage entropy            | 1.69×         | High                             |
| High entropy writes $\geq 0.90$ | 7.98×         | Very High                        |
| Average memory read activity    | 3.67×         | Moderate                         |
| Average memory write activity   | 4.35×         | Moderate                         |
| Storage-memory correlation      | 0.632         | Moderate                         |
| Peak ATA write activity         | 12.71×        | Very High                        |
| High-write/high-entropy windows | 11 seconds    | High-value investigation windows |

```

        timeline_dfs["ata_write.csv"]
        .set_index("timestamp")
        .resample("1s")
        .size()
        .max()
    )

    idle_ata_peak = (
        idle_dfs["ata_write.csv"]
        .set_index("timestamp")
        .resample("1s")
        .size()
        .max()
    )

    ata_peak_ratio = conti_ata_peak / idle_ata_peak

    evidence.append({
        "Forensic Indicator": "Peak ATA write activity",
        "Conti/Idle": ata_peak_ratio,
        "Evidence Strength": "VERY HIGH",
        "Observed Evidence":
            f"{conti_ata_peak:,} vs "
            f"{idle_ata_peak:,} events/sec"
    })

    # -----
    # FINAL TABLE
    # -----

    evidence_df = pd.DataFrame(evidence)

    print("\nFINAL EVIDENCE MATRIX")
    print("=" * 100)

    display(
        evidence_df.round(3)
    )

```

=====

FINAL FORENSIC EVIDENCE ASSESSMENT – CONTI vs IDLE

=====

FINAL EVIDENCE MATRIX

=====

|   | Forensic Indicator                  | Conti/Idle | Evidence Strength | Observed Evidence            |
|---|-------------------------------------|------------|-------------------|------------------------------|
| 0 | Storage write volume                | 109.133    | VERY HIGH         | 2108.16 MB vs 19.32 MB       |
| 1 | Storage write event frequency       | 110.919    | VERY HIGH         | 554,371 vs 4,998 events      |
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| 5 | Average memory-read activity        | 3.674      | MODERATE          | 3025.08 vs 823.37 events/sec |
| 6 | Average memory-write activity       | 4.348      | MODERATE          | 3509.04 vs 807.05 events/sec |
| 7 | Storage-memory temporal correlation | 0.632      | MODERATE          | Correlation = 0.632          |
| 8 | Peak ATA write activity             | 12.714     | VERY HIGH         | 12,434 vs 978 events/sec     |

Figure 10. Final indicator summary produced from the forensic notebook.

## References

1. Hirano et al., “RanSMAP: Open dataset of Ransomware Storage and Memory Access Patterns,” Computers & Security, 2025.
2. RanSMAP dataset, Ransomware Storage and Memory Access Patterns. Dataset source identified in the case-study materials.
3. ContiForensics.ipynb - analysis notebook containing the data preparation, forensic calculations, comparative analysis, correlation analysis, high-intensity window identification, and final evidence matrices.

## Appendix A — Evidence Snapshot Index

| Figure    | Evidence represented                         | Primary report section |
|-----------|----------------------------------------------|------------------------|
| Figure 1  | Selected Conti execution and six artifacts   | 3–4                    |
| Figure 2  | Storage-write activity and entropy over time | 6–8                    |
| Figure 3  | Relative Conti/Idle storage comparison       | 7                      |
| Figure 4  | Conti vs Idle memory comparison              | 9                      |
| Figure 5  | Cross-artifact storage/memory activity       | 10                     |
| Figure 6  | 11 high-write/high-entropy windows           | 11                     |
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| Figure 8  | Peak-activity caveat                         | 13                     |
| Figure 9  | Final forensic assessment                    | 14                     |
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The screenshots are included as evidentiary snapshots of the executed notebook outputs. The notebook remains the authoritative reproducibility artifact for the calculations.